



A Note on Production Inventories

The Purpose of Inventory

Wherever they appear, inventories serve one basic purpose: they allow the rate at which goods are produced or acquired to differ from the rate at which they are required, demanded or used. In different contexts, the reasons for wanting to disconnect the rates of production and consumption may vary considerably, but the basic purpose remains the same: to trade-off the costs of carrying inventories against economies or efficiencies obtained elsewhere. Examples include:

1. Purchasing raw materials in larger quantities and at different times than required by production in order to obtain purchase discounts. (Inventory carrying costs traded off against price breaks.)
2. Using work-in-process inventories to allow an expensive machine with low hourly capacity to produce its daily quota by running round the clock, while a preceding or successive labor intensive stage produces its quota by operating at a higher rate during normal work hours. (This trade-off will thus involve considering the costs of overtime, extra machine capacity and the costs of carrying inventory.)
3. Using finished goods inventories to allow a steady production rate rather than adjusting production rate during the year to match seasonal demands. (This will involve trading off inventory carrying costs against the costs of adjusting the production rate, such as overtime, temporary hiring and lay-offs, training costs, maintaining capacity to match the peak rate, etc.)
4. Keeping inventories of finished goods above the level of demand forecasted in order to avoid the risk of stocking out or expediting production. (This will involve trading off carrying costs against lost sales, loss of goodwill, expediting costs, etc.)
5. Producing goods on one machine, in larger quantities (and with less frequency) than the pattern of demand in order to avoid incurring frequent “set-up” costs or changeover costs involved in switching production from one item to another. (Trading off inventory costs against downtime and direct labor involved in set-ups and changeovers.)

This note was prepared by Associate Professor David H. Maister, as the basis for class discussion rather than to illustrate either effective or ineffective handling of an administrative situation.

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These examples illustrate that, while the principle of “inventory as buffer” is a common theme, the considerations involved in different inventory contexts may vary significantly. The cost items to be examined in deciding whether to use inventories (and if so, how much) need to be examined carefully on a case-by-case basis. As shown above, inventories can be used to save on purchase costs, avoid disruptions between successive stages in a process, reduce peak capacity requirements, avoid overtime and training costs, avoid machine and worker downtime, avoid stockout costs, expediting costs and many other types of costs.

On the other hand, carrying inventories is not cheap. Inventories tie up working capital funds, require space, handling and run the risks of damage and obsolescence. The costs of carrying inventory are normally calculated by discovering the *average* amount of inventory (in units) carried over a period of time (say a year), calculating the value of this inventory using a cost per unit and then assigning a “carrying rate” charge to this dollar amount. In 1983, carrying charges of 20 to 30 percent per year are quite common. Thus, if there is an average inventory (over the year) of 10,000 units of a \$5 item and a 25% rate is applied, the inventory carrying cost for the year will be:

$$(10,000) \times (\$5) \times (0.25) = \$12,500$$

Controlling Inventory Levels

When producing goods to inventory, there are two main questions that must be answered:

1. *How much* should we make (or order)? This is the issue of the appropriate LOT SIZE.
2. *When* should we make it (or order it)? This is the issue of the appropriate REORDER POLICY.

These two decisions are not completely independent. To illustrate this, we shall consider the two basic approaches which companies take in making inventory decisions. The first approach, called the REORDER POINT system, operates as follows. Stock levels are watched closely (and continuously) until they fall to some predetermined level (the REORDER POINT) which signals that a new batch of production is required. An order is then placed for a given quantity. The virtue of this system is that the production quantity (the lot size) can be held relatively constant even if demand for the item varies significantly from that forecasted. If demand is less than predicted, the inventories on-hand will fall slowly and will reach the reorder point later than anticipated: production will be postponed. If demand is higher than expected, the reorder point will be reached more quickly and production scheduled earlier. In either event, the lot size can be held reasonably constant.

The second approach to controlling inventories is often termed the PERIODIC REORDER SYSTEM. Under this system inventory levels are not monitored continuously. A fixed review interval is established (say, once a month or once a week) and production (or purchase) orders are placed for whatever amounts are necessary to carry the company through until the next time orders are due to be placed. Orders in this system are more regularly timed, but involve more highly varying production quantities. Whereas the reorder point system deals with uncertain demand by varying the timing of orders, the periodic reorder system deals with uncertain demands by varying the amount ordered.

(A third type of reordering policy is available. The two described so far, reorder point and periodic review, are particularly applicable where demand is uncertain and cannot be forecasted with a high degree of accuracy. However, some environments exist where demands are known with some precision. An example would be the production of component parts to be used in final assembly, where the schedule for assembly operation has already been decided. In this situation we can *calculate*

how many of each component is needed and when it is needed. The reordering of production of the parts can be scheduled appropriately. This approach is known as REQUIREMENTS PLANNING.)

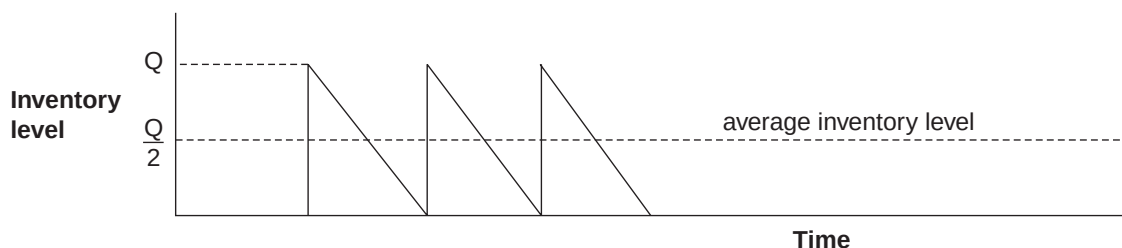
A Look at Lot Sizing

Since inventories can involve trade-offs among a wide variety of different costs, there is no single general procedure for determining the best order quantity or lot size. In many cases, the most advisable way of proceeding is the TOTAL COSTING approach: examine how each cost item varies with orders of different sizes, add up the costs, and plot them on a graph or table, thus discovering for what order size costs are minimized. (Note that in this analysis, only costs that vary with order size need be examined. Other costs may be safely ignored if they remain constant when order size varies.)

However, in certain situations it is possible to use some simple mathematical formulae as a short cut. One formula in particular, which involves some reasonably restrictive assumptions, is very commonly applied. This is the Economic Order Quantity (EOQ) formula.

To introduce this formula, consider the following situation. We have a production facility with a sizeable set-up cost (or changeover cost) of \$800. Whenever we schedule a production run of item X, we incur this fixed set-up cost, regardless of the size of the run. Variable production costs (i.e., cost per item ignoring the set-up cost) at \$4 per unit, are constant regardless of the size of the run. If we choose a large order size, then we will schedule relatively few runs during the year, but carry a large average inventory. If we choose a small production run, we will reduce the average inventory, but incur frequent set-up charges. The inventory carrying rate is 20% per year, and there is a demand for 50,000 units per year.

Assuming that demand (withdrawals from inventory) is reasonably stable, inventory levels over time will look as follows:



When a new production run is received inventories will rise sharply (vertically if the entire run is added to inventory all at once). Inventory levels then fall as withdrawals are made, until they fall to zero and a new production run arrives. (For the moment we will assume that these orders can be perfectly timed.) If the order size is Q , then the average inventory level will be $Q/2$.

A total cost approach in this situation would proceed as follows:

Order Size	5,000	10,000	25,000	50,000
Number of orders per year	10	5	2	1
Set-up cost per year	\$8,000	\$4,000	\$1,600	\$800
Average inventory	2,500	5,000	12,500	25,000
Value of inventory (at \$4 per unit)	10,000	20,000	50,000	100,000
Inventory carrying-cost (at 20% per year)	2,000	4,000	10,000	20,000
Sum of set-up cost and inventory cost	10,000	8,000	11,600	20,800
Variable production costs	200,000	200,000	200,000	200,000
Total costs	210,000	208,000	211,000	220,000

As can be seen, the best action in this situation (among the alternatives explored) is to order 10,000 units five times a year.

This same result could be obtained using the EOQ formula, which has the following expression:

$$EOQ = \sqrt{\frac{2RS}{CK}}$$

Where	R	=	Annual Demand	(50,000)
	S	=	Set-up cost per order	(\$800)
	C	=	Variable cost per unit	(\$4)
	K	=	Inventory carrying charge	(20%)

$$EOQ = \sqrt{\frac{2 \times 50,000 \times 800}{(4 \times 0.20)}}$$

$$= 10,000$$

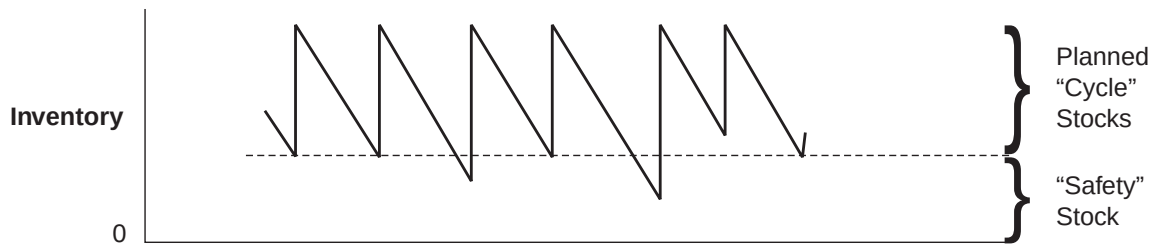
The use of the EOQ formula is a short cut to performing the entire tradeoff analysis shown above, but it should be used with caution. It applies only where:

- i. Demand rate can be forecast with some accuracy and can be assumed to occur at a reasonably steady pace (e.g., without seasonality) at least until the next order. (If the annual demand rate has changed significantly since our last order we will need to recalculate the EOQ.)
- ii. The only relevant costs to be traded off are set-up (or order) costs and inventory carrying costs. If other costs vary with order size (such as variable production costs due to learning, or purchase costs due to discounts) then the EOQ does not apply and a different approach to lot-sizing is required.
- iii. Set-up or order costs are constant regardless of the size of the order.

A Look at Reorder Points

Suppose that we are using a reorder point system to control our inventories, i.e., we place an order whenever sales fall to a predetermined level. What should influence this level? The main influence will be how long it takes for an order to be received from the production facility (or the supplier) once we have placed an order. If this lead time is short, then we can reorder when stocks are running very low. However, if the lead time is long, then we will have to order well in advance of the time we are due to run out. The two main questions of interest are thus (a) How long will it take for an order to be received?; and (b) what amount of demand will take place during this time? The combination of these two will determine the level of inventory at which we place an order for replenishment. It should be noted, however, that there can be uncertainty involved in both the length of the lead time and the rate of demand during lead time. In this case we might set the reorder point even higher. (The precise level would involve a statistical analysis of the probabilities of stocking out, and a trade-off between the costs of this and the costs of carrying the extra “safety” stock.)

With a reorder point set to allow for demand uncertainties during the lead time (and uncertainties in the lead time) the pattern of inventories would look something like this:



Without fluctuations inventories would not fall to zero, but to the preplanned safety stock level. When demand was higher than expected (or lead time longer), inventory would still be available. However, since safety stocks (as shown above) increase the overall level of inventory, they are expensive to maintain on a permanent basis.

A Look at Periodic Reorder Systems

If we decide to control inventories not with a fixed reorder point but by reviewing inventory levels on a fixed regular schedule, we must decide what time interval to use: once a week, once a month, or some other number. One approach to deciding this is to consider the lot sizes (order quantities) that will result. While the lot sizes in a periodic review system might vary from order to order (due to uncertainties in demand that have taken place between reviews, and the consequent need to replenish stocks to a greater or lesser extent), the lot sizes should be reasonably similar unless demand fluctuates extremely. We would still want our lot sizes to represent a good trade-off between the relevant costs. Consequently, we can discover a “good” (if not optimal) order interval by examining what lot sizes we want and then finding out what order interval would give lot sizes approximately this number. For example, if we have demand for a product of 20,000 units per year and an examination of the production system shows that lot sizes of about 600 represent the best trade-off between the various costs involved, then we would want to order approximately $(20,000 / 600)$ or 33 times per year, or once every two days (assuming a 365-day year).

ABC Analysis

Controlling inventories can be an information-intensive, management-intensive and sometimes capital-intensive activity. It involves monitoring activities, checking for inventory record accuracy, potentially frequent updates on demand forecasts, re-examination of costs, and potentially frequent recalculation of decision-rules. In a company with 2,000 line items, each of 5 colors and sizes, there would be 10,000 “stock-keeping units” (SKU’s). It is clear, therefore, that there is a trade-off involved in how much should be spent on the inventory control system compared to the savings available from tightening control. In many situations the costs of tight control can exceed the potential savings. Accordingly, many companies divide their line items (or SKU’s) into A, B, and C categories. A items, the most important, are tracked continually and much management attention given. At the other extreme, C items (least important) are often controlled semi-automatically, with a computer programmed to perform all revisions of forecasts, issuance of orders, and so on. Management attention is brought to bear only on an exception-reporting basis.

The 80-20 rule states that approximately 80 percent of activity comes from 20 percent of the items. While this is only an approximation to most situations, it points out that you should spend greater effort controlling the inventory of a few critical items.